

Curriculum Vitae

Prithish Halder

Department of Physics and Astronomy, Theodore Jorgensen Hall
855 North 16th Street, University of Nebraska-Lincoln, NE 68588-0299, USA
✉ dr.prithishhalder@gmail.com ✉ phalder2@unl.edu  ORCID Profile

Personal Details

Father's Name: Late Goutam Kr. Halder

Mother's Name: Debi Halder

Date of Birth: 06-01-1988

Gender: Male

Marital Status: Married

Home Address: Haral, Champahati,
South 24 Parganas, West Bengal, India

Research & Job Experience

Adjunct Faculty

2024 – Present

Department of Physics & Astronomy, University of Nebraska-Lincoln, USA

Collaborator: Prof. Peter Dowben

Postdoctoral Fellow (RA-III)

2021 – 2023

Indian Institute of Astrophysics, Bangalore, India

Mentor: Prof. Sujan Sengupta

Postdoctoral Fellow (RA-II)

2019 – 2021

Astronomy & Astrophysics Division, Physical Research Laboratory, Ahmedabad, India

Mentor: Prof. Shashikiran Ganesh

Educational Background

PhD in Physics (Dust in Astrophysics)

2013 – 2019

Department of Physics, Assam University, Silchar, India

Thesis: Study of physical properties of cosmic dust from light scattering

Advisor: Prof. Himadri Sekhar Das (Award Date: 1st May 2019)

MSc in Physics

2010 – 2012

Department of Physics, Assam University, Silchar, India

Specialization: Astrophysics

BSc in Physics (Major)

2006 – 2009

Alipurduar College, University of North Bengal, India

Research Interests

- **Dust in Astrophysics:** Comet dust (morphology and composition), Asteroid dust, Interplanetary dust, Proto-planetary dust, Interstellar dust/medium, Circum-galactic dust/medium.
- **Light Scattering Processes:** Polarization, Scattering, Absorption, Extinction.

- **Solar Physics:** Solar energetic particles (SEP), Solar flares (Magnetic topology), Flare Prediction, Stellar flares & CMEs.
- **Atmospheric Science & Nanoscience:** Aerosols, Air borne micro-plastics, Plasmonics, Nano crystals, Perovskite.

Skills & Expertise

- **Programming:** Julia, Fortran, C/C++, Python, R, Bash, Java, Visual Basic, MatLab, $\text{\LaTeX}$
- **Light Scattering Tools:** Mie theory, T-matrix, DDA, RGD
- **Data Reduction & Topology:** IRAF, PyRAF, DS9, DAVE4VM, ARTop, NLFFF, GPU Computing (CUDA.jl)
- **Machine Learning:** XGBoost, CNN, Siamese Neural Network, Random Forest
- **Graphics & Plotting:** Gnuplot, Matplotlib, Origin, VTK, Paraview, Blender 3D

High Performance Computing & Observations

- **Holland Supercomputer** (UNL, USA, 2024): Light scattering, MHD simulation, Solar magnetic topology.
- **NOVA Supercomputer** (IIA, India, 2021): Light scattering of complex comet/cosmic dust.
- **Vikram-100 HPC Center** (PRL, India, 2019): Light scattering of heterogeneous comet dust.
- **Telescopic Observations:** 1.2m Telescope at Mt. Abu Infrared Observatory (PRL, 2020) and 1.04m Sampurnanand Telescope (ARIES, 2016) for Comets and Molecular clouds.

Awards, Fellowships & Achievements

- **Seal of Excellence Award** (Horizon Europe, MSCA, Brussels): 2025, 2026
- **Best Poster Award** (Indian Academy of Sciences): 39th ASI Meeting, 2021
- **Fellowships:** IIA Regular Postdoctoral Fellowship (2021), PRL Postdoctoral Fellowship (2019), UGC Non-NET Fellowship (2013)

Memberships & Peer Reviews

- **Memberships:** Astronomical Society of India (Life Member, 2022), International Astronomical Union (Junior Member, 2021)
- **Peer Reviewer for:** Journal of Quantitative Spectroscopy & Radiative Transfer, New Astronomy, Scientific Reports, Philosophical Transactions A

Software Development

- **FastTOP (Fast TOPology extraction for solar active regions):** (2026) developed in Julia and CUDA.jl.
- **REST (Rough Ellipsoid Structure Tools):** (2022) developed in Fortran, Python, and VTK.
- **JaSTA (Java Superposition T-matrix Application):** Versions 1.0, 2.0, 3.0 (2014 – 2021) developed in Fortran and Java for light scattering of aggregates.

Workshops, Conferences, Talks & Posters

- **Talk:** Prediction of solar flare occurrence and its precise location through magnetic topological parameters, *44th ASI Meeting*, IIT Guwahati, India (2026).
- **Invited Talk:** Physics-based Flare Forecasting: Role of Winding Flux and Persistent PILS Evolution During Precursor Phase of the Intense Solar Flares, *ASIS003*, JECRC University, Jaipur, India (2025).
- **Invited Talk:** Recent Trends of Machine Learning in Astronomy on National Space Day 2024, Sanjivani University, Pune, India (2024).
- **Invited Talk:** Comet & recent trends in cometary studies, Sunbeam Women's College Varuna, Varanasi, India (2022).
- **Invited Talk:** Modelling comet dust particles: morphology and light scattering, *IIA Foundation Day Symposium*, IIA Bangalore, India (2022).
- **Poster:** Crafting cosmic dust from ellipsoids, *40th ASI Meeting*, ARIES Nainital & IIT Roorkee, India (2022).
- **Poster:** A comet dust model based on the findings from the Rosetta mission, *39th ASI Meeting*, ICTS - TIFR Bengaluru, India (2021).
- **Invited Talk:** Classification of dust particles and their polarimetric response, *ICDA-2020*, Assam University, Silchar, India (2020).
- **Thesis Presentation:** Study of physical properties of cosmic dust from light scattering, *38th ASI Meeting*, IISER Tirupati, India (2020).
- **Invited Talk:** Dust Morphology in Comets and Proto-planetary Disk, *Planet Formation Workshop*, NAOJ, Tokyo, Japan (2019).
- **Oral:** Computational challenges in modeling the morphologically complex dust particles, *Conference on Infrared Astronomy and Astrophysical Dust*, IUCAA Pune, India (2019).
- **Invited Talk:** Investigating the effect of porosity on the light scattering properties of aggregated dust particles, *NEMA-II*, Tezpur University, India (2016).

Research Metrics

- **Google Scholar:** Citations: 132 | h-index: 7 | [Profile Link](#)
- **ResearchGate:** Citations: 137 | h-index: 7 | Research Interest: 118 | [Profile Link](#)

Publications

- [25] G. Ahuja, S. Ganesh, K. Aravind, E. Jehin, B. Ailawadhi, V. K. Agnihotri, B. Arvind, P. Saraf, A. Singh, D. R. Sarkar, S. Chandra, **P. Halder**, D. Sahu, T. Sivarani, and A. Bhardwaj. “Post-perihelion observations of the interstellar comet 3I/ATLAS”. In: *Monthly Notices of the Royal Astronomical Society* (2026). “(Accepted)”.
- [24] M. Arnaut, C. Wöhler, **P. Halder**, G. Ahuja, S. Ganesh, and M. Bhatt. “Wide-field polarization imaging and numerical modeling of the coma and tail of comet C/2023 A3 (Tsuchinshan-ATLAS)”. In: *A&A* 707, A150 (Mar. 2026), A150. doi: 10.1051/0004-6361/202555905. arXiv: 2506.21284 [astro-ph.EP].
- [23] S. K. Pal, D. Sen, D. Bardhan, **P. Halder**, and S. K. Ghosh. “Optical Properties of One Dimensional Plasmonic Polymers”. In: *ChemPhysChem* 27 (2 Jan. 2026), e202500862. doi: 10.1002/CPHC.202500862. URL: /doi/pdf/10.1002/cphc.202500862%20https://onlinelibrary.wiley.com/doi/abs/10.1002/cphc.202500862%20https://chemistry-europe.onlinelibrary.wiley.com/doi/abs/10.1002/cphc.202500862.
- [22] D. Debnath, **P. Halder**, B. Saha, M. Das, and S. K. Ghosh. “Exploring the Plasmonic Density of States To Direct Singlet or Triplet Excitonic Energy Transfer from Perovskite Nanocrystals”. In: *The Journal of Physical Chemistry C* 129 (13 Apr. 2025), pp. 6535–6550. doi: 10.1021/ACS.JPCC.5C01599. URL: https://pubs.acs.org/doi/abs/10.1021/acs.jpcc.5c01599.
- [21] S. Narendranath, S. Ganesh, D. Sahu, R. K. Mishra, M. Bhatt, S. Sarkhel, G. Thangjam, K. K. Marhas, K. Aravind, K. Venkataramani, **P. Halder**, P. Prajapati, and P. Sreekumar. “Solar system research prospects for the decade and beyond”. In: *Journal of Astrophysics and Astronomy* 46.2, 34 (June 2025), p. 34. doi: 10.1007/s12036-025-10060-0.
- [20] D. Debnath, **P. Halder**, B. Saha, H. Acharya, and S. K. Ghosh. “Fermi–Dirac statistics in correlation between electronic entropy and spatial confinement of liquid crystal-perovskite nanohybrids”. In: *Journal of Materials Chemistry C* 12.10 (Mar. 2024), pp. 3535–3548. doi: 10.1039/d4tc00212a. eprint: https://pubs.rsc.org/tc/article-pdf/12/10/3535/10160782/d4tc00212a.pdf. URL: https://doi.org/10.1039/d4tc00212a.
- [19] B. Goswami, H. S. Das, N. N. Devi, and **P. Halder**. “Influence of porosity on the Umov effect in silicate and organic refractory aggregates”. In: *JQSRT* 323, 109019 (Sept. 2024), p. 109019. doi: 10.1016/j.jqsrt.2024.109019.
- [18] R. S. Paul, **P. Halder**, H. S. Das, A. M. Mazarbhuiya, and B. J. Medhi. “Investigating the extinction and magnetic field geometry in the Bok globule CB26”. In: *MNRAS* 535.4 (Dec. 2024), pp. 3353–3362. doi: 10.1093/mnras/stae2497.
- [17] **P. Halder** and S. Sengupta. “A Comprehensive Model of Morphologically Realistic Cosmic Dust Particles: An Application to Mimic the Unusual Polarization Properties of the Interstellar Comet 2I/Borisov”. In: *ApJ* 947.1, 1 (Apr. 2023), p. 1. doi: 10.3847/1538-4357/acbf52. arXiv: 2302.13370 [astro-ph.EP].
- [16] K. Aravind, **P. Halder**, S. Ganesh, D. Sahu, M. Serra-Ricart, J. J. Chambó, D. Angchuk, and T. Sivarani. “Optical observations and dust modelling of comet 156P/Russell-LINEAR”. In: *Icarus* 383, 115042 (Sept. 2022), p. 115042. doi: 10.1016/j.icarus.2022.115042. arXiv: 2204.09727 [astro-ph.EP].
- [15] **P. Halder**. “REST: A Java Package for Crafting Realistic Cosmic Dust Particles”. In: *ApJS* 263.1, 3 (Nov. 2022), p. 3. doi: 10.3847/1538-4365/ac9183. arXiv: 2209.05768 [astro-ph.EP].

- [14] A. M. Mazarbhuiya, H. S. Das, B. J. Medhi, **P. Halder**, and P. Deb Roy. “Study of dust coma of comets 32P/Comas Sola and C/2015 V2 (Johnson) by imaging polarimetry”. In: *Ap&SS* 367.9, 98 (Sept. 2022), p. 98. doi: 10.1007/s10509-022-04133-7.
- [13] **P. Halder**. “JaSTA-3: Light scattering simulations for heterogeneous aggregate”. In: *Computer Physics Communications* 261, 107818 (Apr. 2021), p. 107818. doi: 10.1016/j.cpc.2020.107818.
- [12] **P. Halder** and S. Ganesh. “Modelling heterogeneous dust particles: an application to cometary polarization”. In: *MNRAS* 501.2 (Feb. 2021), pp. 1766–1781. doi: 10.1093/mnras/staa3647. arXiv: 2011.08252 [astro-ph.EP].
- [11] A. M. Mazarbhuiya, H. S. Das, and **P. Halder**. “The Umov effect in cosmic dust analogue fluffy aggregates”. In: *MNRAS* 502.2 (Apr. 2021), pp. 2536–2540. doi: 10.1093/mnras/stab230.
- [10] **P. Halder**, P. Deb Roy, and H. S. Das. “Dependence of light scattering properties on porosity, size and composition of dust aggregates”. In: *Icarus* 312 (Sept. 2018), pp. 45–60. doi: 10.1016/j.icarus.2018.04.026. arXiv: 1804.08324 [astro-ph.EP].
- [9] P. Deb Roy, **P. Halder**, and H. S. Das. “Study of light scattering properties of dust aggregates with a wide variation of porosity”. In: *Ap&SS* 362.11, 209 (Nov. 2017), p. 209. doi: 10.1007/s10509-017-3185-x. arXiv: 1709.09319 [astro-ph.EP].
- [8] **P. Halder** and H. S. Das. “JaSTA-2: Second version of the Java Superposition T-matrix Application”. In: *Computer Physics Communications* 221 (Dec. 2017), pp. 421–422. doi: 10.1016/j.cpc.2017.08.020.
- [7] P. Deb Roy, **P. Halder**, H. S. Das, and B. J. Medhi. “Imaging polarimetry of comets C/2013 V1 (Boattini) and 290P/Jager before and after perihelion”. In: *MNRAS* 450.2 (June 2015), pp. 1770–1776. doi: 10.1093/mnras/stv707.
- [6] P. Deb Roy, **P. Halder**, H. S. Das, and B. J. Medhi. “Polarimetric Studies of comets C/2013 V1 (Boattini) and 290P/Jager”. In: *IAU General Assembly*. Vol. 29. Aug. 2015, 2256508, p. 2256508.
- [5] **P. Halder**, A. Chakraborty, P. Deb Roy, and H. S. Das. “Java application for the superposition T-matrix code to study the optical properties of cosmic dust aggregates”. In: *Computer Physics Communications* 185.9 (Sept. 2014), pp. 2369–2379. doi: 10.1016/j.cpc.2014.04.019. arXiv: 1405.7306 [physics.comp-ph].

Publications Under Review

- [4] A. Dalal, **P. Halder**, and J. P. DeLong. “Temperature-dependent risk cues suppress the functional response of copepods”. In: *Freshwater Biology* (2026). “(Under Review)”.
- [3] **P. Halder**, S. K. Mishra, A. K. Srivastava, A. Asai, and P. A. Dowben. “Magnetic Topological Evolution and the Predictive Role of Winding-flux-based Masked PILs in Extreme Solar Flares from AR13664 and AR13842”. In: *ApJS* (2026). “(Under Review)”.
- [2] **P. Halder**, S. K. Mishra, A. K. Srivastava, and P. A. Dowben. “FastTOP: A scalable GPU framework for accelerated magnetic topology analysis of solar active regions”. In: *Nature Computational Sciences* (2026). “(Under review)”.
- [1] S. Mishra, A. Asai, **P. Halder**, S. Mondal, and A. K. Srivastava. “Successive Emerging Flux Triggers Persistent Reconnection Jets in the Solar Atmosphere and Associated Heating”. In: *ApJ* (2026). “(Under Review)”.